Ge 158 Lecture 5: EM Wave Transmission and Reflection at Planar Interfaces, Part 2: Oblique Incidence

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Topics covered

- Reflection and transmission at oblique incidence

1 EM boundary conditions (review)

The EM boundary conditions (BCs) for zero charge and zero current at the interface, in lossless media (Lecture 4), with $\hat{\mathbf{a}}$ the unit normal to the interface, are:

1. $\varepsilon_1' E_1^\perp = \varepsilon_2' E_2^\perp$ or $\hat{\mathbf{a}} \cdot (\varepsilon_1' \mathbf{E}_1 - \varepsilon_2' \mathbf{E}_2) = 0$

   $\rightarrow$ the normal component of $\varepsilon' \mathbf{E}$ is continuous;

2. $\mu_1 H_1^\perp = \mu_2 H_2^\perp$ or $\hat{\mathbf{a}} \cdot (\mu_1 \mathbf{H}_1 - \mu_2 \mathbf{H}_2) = 0$

   $\rightarrow$ the normal component of $\mu \mathbf{H}$ is continuous;

3. $\mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel$ or $\hat{\mathbf{a}} \times (\mathbf{E}_1 - \mathbf{E}_2) = \mathbf{0}$

   $\rightarrow$ the tangential component of $\mathbf{E}$ is continuous;

4. $\mathbf{H}_1^\parallel = \mathbf{H}_2^\parallel$ or $\hat{\mathbf{a}} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{0}$

   $\rightarrow$ the tangential component of $\mathbf{H}$ is continuous.

2 Reflection and transmission at oblique incidence: geometric optics

Let's start with the basics: "geometric optics" (Figure 1).

2.1 Fields for a uniform plane wave

With position vector $\mathbf{r}$, the fields of the incident, reflected, and transmitted uniform plane waves are, in terms of phasors [$\mathbf{E}^i(\mathbf{r}, t) = \mathrm{Re}\{\tilde{\mathbf{E}}^i(\mathbf{r})\, e^{j\omega t}\}$, etc.],

$$\text{incident:} \quad \tilde{\mathbf{E}}^i(\mathbf{r}) = \tilde{\mathbf{E}}^i(\mathbf{0})\, e^{-j\mathbf{k}_i \cdot \mathbf{r}}, \qquad \tilde{\mathbf{H}}^i(\mathbf{r}) = \frac{1}{\eta_1} \left(\hat{\mathbf{k}}_i \times \tilde{\mathbf{E}}^i(\mathbf{r}) \right), \tag{1}$$

$$\text{reflected:} \quad \tilde{\mathbf{E}}^r(\mathbf{r}) = \tilde{\mathbf{E}}^r(\mathbf{0})\, e^{-j\mathbf{k}_r \cdot \mathbf{r}}, \qquad \tilde{\mathbf{H}}^r(\mathbf{r}) = \frac{1}{\eta_1} \left(\hat{\mathbf{k}}_r \times \tilde{\mathbf{E}}^r(\mathbf{r}) \right), \tag{2}$$

$$\text{transmitted:} \quad \tilde{\mathbf{E}}^t(\mathbf{r}) = \tilde{\mathbf{E}}^t(\mathbf{0})\, e^{-j\mathbf{k}_t \cdot \mathbf{r}}, \qquad \tilde{\mathbf{H}}^t(\mathbf{r}) = \frac{1}{\eta_2} \left(\hat{\mathbf{k}}_t \times \tilde{\mathbf{E}}^t(\mathbf{r}) \right), \tag{3}$$

where $\hat{\mathbf{k}}_i = \mathbf{k}_i / k_i$, etc., are the propagation unit vectors (Lecture 2).

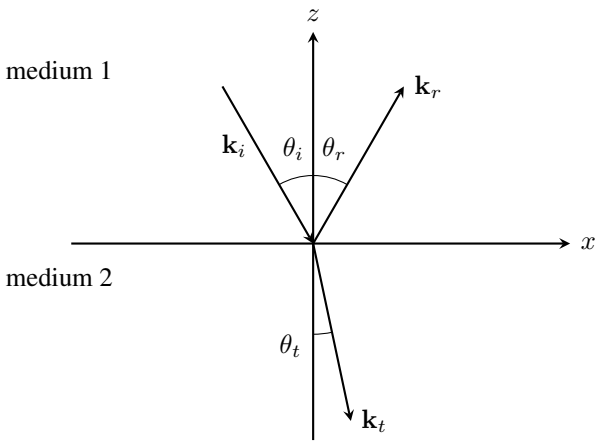


Figure 1: Geometry of reflection and transmission at oblique incidence on the planar interface ($z = 0$) between medium 1 and medium 2. $\mathbf{k}_i$ is the wavenumber vector of the incident wave (and similarly $\mathbf{k}_r$ and $\mathbf{k}_t$ for the reflected and transmitted waves).

- Recall that ω is the same everywhere and for all time and is set by the source.

  $\Rightarrow$ the dispersion relation gives

$$\omega = k_i u_{p1} = k_r u_{p1} = k_t u_{p2} \tag{4}$$

$$\Rightarrow \quad \boxed{k_i = k_r = \frac{u_{p2}}{u_{p1}}\, k_t = \frac{\mathrm{Re}\left(\sqrt{\varepsilon_1}\right)}{\mathrm{Re}\left(\sqrt{\varepsilon_2}\right)}\, k_t = \frac{n_1'}{n_2'}\, k_t} \tag{5}$$

  (lossless media; see Section 4.1 for lossy media).

- The total $\mathbf{E}$ and $\mathbf{H}$ fields in medium 1 are related to those in medium 2 through the BCs, which have the generic structure

$$B^i\, e^{j(\omega t - \mathbf{k}_i \cdot \mathbf{r})} + B^r\, e^{j(\omega t - \mathbf{k}_r \cdot \mathbf{r})} = B^t\, e^{j(\omega t - \mathbf{k}_t \cdot \mathbf{r})}. \tag{6}$$

  Don't worry about the B coefficients for now; just note that the only space and time dependence is in the exponents.

2.2 Matching the exponents at the interface

- The BCs must hold everywhere on the xy plane at the interface $z = 0$ for all times.

  $\Rightarrow$ the exponential factors in Eq. (6) must be equal at $z = 0$:

  - the time factors are already equal because $\omega = \text{const}$;
  - the spatial components give

$$\mathbf{k}_i \cdot \mathbf{r} = \mathbf{k}_r \cdot \mathbf{r} = \mathbf{k}_t \cdot \mathbf{r} \qquad \text{at } z = 0. \tag{7}$$

    Expanding,

$$x k_{ix} + y k_{iy} = x k_{rx} + y k_{ry} = x k_{tx} + y k_{ty} \tag{8}$$

    for all x and y $\Rightarrow$ this only holds if the components are separately equal:

    $\hookrightarrow$ for $x = 0$: $\quad k_{iy} = k_{ry} = k_{ty}$;

$\hookrightarrow$ for $y = 0$: $\quad k_{ix} = k_{rx} = k_{tx}$.

- We are free to choose our coordinate system; for convenience, let's define $\mathbf{k}_i$ to lie in the x–z plane

$$\Rightarrow \quad k_{iy} = k_{ry} = k_{ty} = 0, \tag{9}$$

and the only non-zero components of the wavenumber vectors lie in the x–z plane for the incident, reflected, and transmitted waves.

2.3 The three laws of geometric optics

From the above, we get

1st law of geometric optics: The incident, reflected, and transmitted wave vectors (rays) form a plane called the *plane of incidence*, which also includes the normal to the surface (in our case $\hat{\mathbf{z}}$).

Now, given that $k_{ix} = k_{rx} = k_{tx}$,

$$k_i \sin\theta_i = k_r \sin\theta_r = k_t \sin\theta_t, \tag{10}$$

where, all measured with respect to the normal,

- $\theta_i \equiv$ angle of incidence;
- $\theta_r \equiv$ angle of reflection;
- $\theta_t \equiv$ angle of refraction (or transmission).

From the dispersion relation, we have $k_i = k_r$, so

$$\boxed{\theta_i = \theta_r \equiv \theta_1} \tag{11}$$

2nd law of geometric optics, commonly called the *law of reflection*: angle of incidence θ_i = angle of reflection θ_r.

And from the dispersion relation, we have [Eq. (5)]

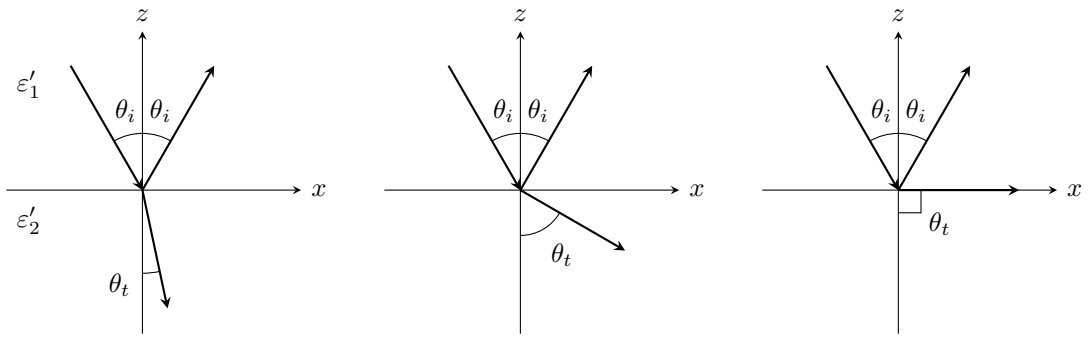
$$k_i = k_r = k_1 = \frac{u_{p2}}{u_{p1}}\, k_t = \frac{\mathrm{Re}\left(\sqrt{\varepsilon_1}\right)}{\mathrm{Re}\left(\sqrt{\varepsilon_2}\right)}\, k_t = \frac{n_1'}{n_2'}\, k_t. \tag{12}$$

3rd law of geometric optics, a.k.a. the *law of refraction* or *Snell's law* (lossless media):

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{u_{p2}}{u_{p1}} = \frac{n_1'}{n_2'} = \frac{\mathrm{Re}\left(\sqrt{\varepsilon_1}\right)}{\mathrm{Re}\left(\sqrt{\varepsilon_2}\right)} \tag{13}$$

- Note that up to this point, we did not invoke any EM principles to derive the three laws of geometric optics.

  $\Rightarrow$ These laws are valid for all wave types (water, sound, seismic, etc.), and many of you will have seen them before.



(a) $\varepsilon_1' < \varepsilon_2' \Rightarrow \theta_t < \theta_i$ "inward refraction"

(b) $\varepsilon_1' > \varepsilon_2' \Rightarrow \theta_t > \theta_i$ "outward refraction"

(c) $\theta_i = \theta_c$ (critical angle) $\Rightarrow \theta_t = \pi/2$

Figure 2: Implications of Snell's law for lossless media: (a) inward refraction ($\varepsilon_1' < \varepsilon_2'$); (b) outward refraction ($\varepsilon_1' > \varepsilon_2'$); (c) incidence at the critical angle θ_c, where the refracted wave travels along the boundary.

2.4 Implications of Snell's law

Before moving on to EM waves, let's quickly consider the implications of Snell's law. Let's keep it simple and consider lossless media (Figure 2).

- $\varepsilon_1' < \varepsilon_2' \Rightarrow \theta_t < \theta_i$: "inward refraction" (Figure 2a).
- $\varepsilon_1' > \varepsilon_2' \Rightarrow \theta_t > \theta_i$: "outward refraction" (Figure 2b).
- *Critical angle:* $\theta_i = \theta_c \rightarrow \theta_t = \pi/2$ (Figure 2c). To find θ_c, apply Snell's law:

$$\sin\theta_c = \frac{n_2'}{n_1'}\sin\theta_t\bigg|_{\theta_t=\pi/2} = \frac{n_2'}{n_1'} \quad \Rightarrow \quad \theta_c = \sin^{-1}\left(\frac{\mathrm{Re}\left(\sqrt{\varepsilon_2}\right)}{\mathrm{Re}\left(\sqrt{\varepsilon_1}\right)}\right). \tag{14}$$

  - When $\theta_i = \theta_c$, the refracted wave travels along the boundary.
  - When $\theta_i > \theta_c$: total reflection (only possible when $n_1' > n_2'$).
    $\hookrightarrow$ gives diamonds sparkle.

3 Reflection and transmission of EM waves at oblique incidence

Now that we have an okay grasp of the exponential terms, let's return to EM waves.

3.1 Applying the boundary conditions

Applying the BCs discussed before to the components of the field phasors at the origin (the exponential factors are equal everywhere on $z = 0$):

1. Gauss's law:
$$\varepsilon_1'\left(\tilde{E}_z^i + \tilde{E}_z^r\right) = \varepsilon_2'\tilde{E}_z^t; \tag{15}$$
2. Gauss's law for magnetism:
$$\mu_1\left(\tilde{H}_z^i + \tilde{H}_z^r\right) = \mu_2\tilde{H}_z^t; \tag{16}$$

3. Faraday's law:

$$\tilde{E}_x^i + \tilde{E}_x^r = \tilde{E}_x^t, \qquad \tilde{E}_y^i + \tilde{E}_y^r = \tilde{E}_y^t; \tag{17}$$

4. Ampère's law:

$$\tilde{H}_x^i + \tilde{H}_x^r = \tilde{H}_x^t, \qquad \tilde{H}_y^i + \tilde{H}_y^r = \tilde{H}_y^t. \tag{18}$$

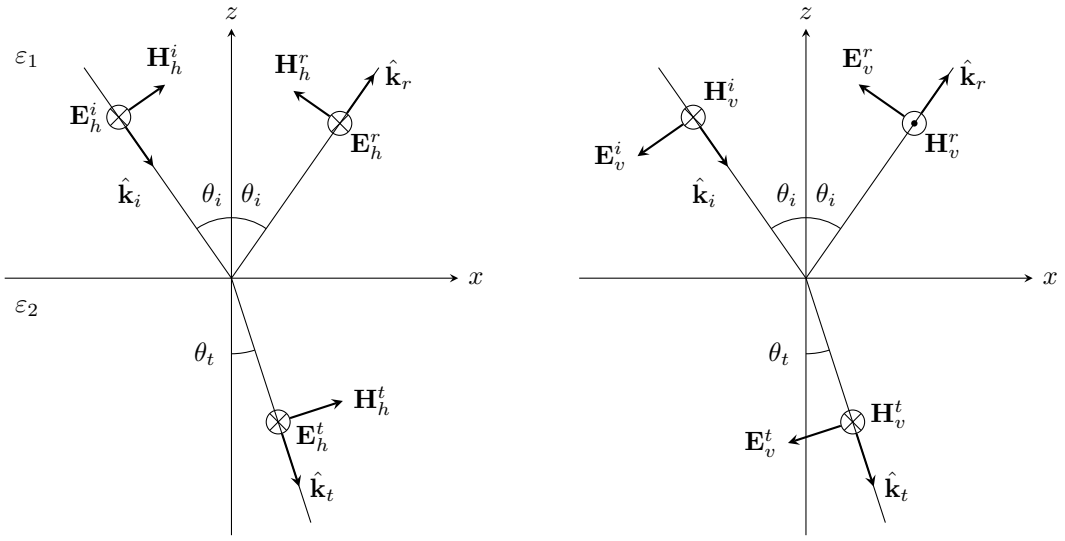
3.2 Polarization definitions

Now we need to get organized by defining polarizations.

- What's a good reference for polarization? The plane of incidence (see the 1st law of geometric optics).

  $\Rightarrow$ two commonly defined polarizations (Figure 3):

  1. *perpendicular* or *horizontal* $\rightarrow$ the $\mathbf{E}$ field is normal to the plane of incidence;
  2. *parallel* or *vertical* $\rightarrow$ the $\mathbf{H}$ field is normal to the plane of incidence (so named because the $\mathbf{E}$ field lies in the plane of incidence and usually has a vertical component).



(a) perpendicular or horizontal (H) (b) parallel or vertical (V)

Figure 3: Reference directions of the fields for (a) perpendicular or horizontal (H) and (b) parallel or vertical (V) polarization. $\otimes$: into the page ($+\hat{\mathbf{y}}$); $\odot$: out of the page ($-\hat{\mathbf{y}}$). The plane of incidence is the x–z plane (the page). In (b), the reflected-wave reference directions are those for which ρ_v and τ_v take the forms in Eqs. (32) and (33).

- Why does $\mathbf{H}_v$ flip on reflection in Figure 3b? The fields and the propagation direction always form a right-handed set, with $\mathbf{E} \times \mathbf{H}$ along $\hat{\mathbf{k}}$. The reflected wave travels upward ($\hat{\mathbf{k}}_r$ has a $+\hat{\mathbf{z}}$ component), while the incident wave travels downward, so when $\hat{\mathbf{k}}$ reverses, either $\mathbf{E}$ or $\mathbf{H}$ must reverse too. The reflected $\mathbf{E}_v^r$ reference is chosen to match $\mathbf{E}_v^i$ at normal incidence, so $\mathbf{H}_v^r$ is the one that flips. At $\theta_i = 0$, with $\mathbf{H} = \hat{\mathbf{k}} \times \mathbf{E}/\eta$:

  – incident: $\hat{\mathbf{k}}_i = -\hat{\mathbf{z}}$ and $\mathbf{E}_v^i \parallel -\hat{\mathbf{x}}$, so $\mathbf{H}_v^i \parallel (-\hat{\mathbf{z}}) \times (-\hat{\mathbf{x}}) = +\hat{\mathbf{y}}$ ($\otimes$);

- reflected: $\hat{\mathbf{k}}_r = +\hat{\mathbf{z}}$ and $\mathbf{E}_v^r \parallel -\hat{\mathbf{x}}$, so $\mathbf{H}_v^r \parallel (+\hat{\mathbf{z}}) \times (-\hat{\mathbf{x}}) = -\hat{\mathbf{y}}$ ($\odot$);
- transmitted: $\hat{\mathbf{k}}_t$ still points downward, so $\mathbf{H}_v^t$ keeps the incident direction ($\otimes$).

· The same thing happens for H polarization (Figure 3a), where $\mathbf{E}_h$ is $\otimes$ for all three waves and the in-plane $\mathbf{H}_h$ reverses instead: at normal incidence, $\mathbf{H}_h^i \parallel +\hat{\mathbf{x}}$ and $\mathbf{H}_h^r \parallel -\hat{\mathbf{x}}$. The flip is easier to see for V polarization because the field normal to the page is $\mathbf{H}$.

· The arrows are reference directions, not the actual field directions: the actual reflected field is the reference direction times the (complex) amplitude, $E_v^r = \rho_v E_v^i$. For example, at normal incidence onto a denser medium ($\eta_2 < \eta_1$), $\rho_v < 0$, so the actual $\mathbf{E}_v^r$ points opposite its arrow and the actual $\mathbf{H}_v^r$ points into the page. With these reference directions, $\rho_v = \rho_h = \rho$ at normal incidence, as in Lecture 4.

· As we discussed in the lecture on polarization, any polarization can be decomposed into orthogonal linear polarizations.

⇒ Hereafter, we only need to consider horizontal (H) and vertical (V) polarizations (the same as perpendicular and parallel, respectively).

3.3 H polarization

The fields are

$$\tilde{\mathbf{E}}_h^i = \hat{\mathbf{y}}\, E_h^i\, e^{-\gamma_1(x\sin\theta_i - z\cos\theta_i)}, \quad \tilde{\mathbf{H}}_h^i = (\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{z}}\sin\theta_i)\,\frac{E_h^i}{\eta_1}\, e^{-\gamma_1(x\sin\theta_i - z\cos\theta_i)} \quad \text{(incident)}, \tag{19}$$

$$\tilde{\mathbf{E}}_h^r = \hat{\mathbf{y}}\, E_h^r\, e^{-\gamma_1(x\sin\theta_i + z\cos\theta_i)}, \quad \tilde{\mathbf{H}}_h^r = (-\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{z}}\sin\theta_i)\,\frac{E_h^r}{\eta_1}\, e^{-\gamma_1(x\sin\theta_i + z\cos\theta_i)} \quad \text{(reflected)}, \tag{20}$$

$$\tilde{\mathbf{E}}_h^t = \hat{\mathbf{y}}\, E_h^t\, e^{-\gamma_2(x\sin\theta_t - z\cos\theta_t)}, \quad \tilde{\mathbf{H}}_h^t = (\hat{\mathbf{x}}\cos\theta_t + \hat{\mathbf{z}}\sin\theta_t)\,\frac{E_h^t}{\eta_2}\, e^{-\gamma_2(x\sin\theta_t - z\cos\theta_t)} \quad \text{(transmitted)}. \tag{21}$$

3.3.1 Applying the BCs

· From Gauss's law, Eq. (15),

$$\varepsilon_1' \left(\tilde{E}_z^i + \tilde{E}_z^r\right) = \varepsilon_2' \tilde{E}_z^t, \tag{22}$$

but $\tilde{E}_z^i = \tilde{E}_z^r = \tilde{E}_z^t = 0$, so there is no information.

· From Gauss's law for magnetism, Eq. (16), with $\mu_1 = \mu_2 = \mu_0$,

$$\tilde{H}_z^i + \tilde{H}_z^r = \tilde{H}_z^t, \tag{23}$$

$$\frac{E_h^i}{\eta_1}\sin\theta_i + \frac{E_h^r}{\eta_1}\sin\theta_i = \frac{E_h^t}{\eta_2}\sin\theta_t. \tag{24}$$

Applying Snell's law and canceling terms, we get

$$\underbrace{E_h^i + E_h^r = E_h^t}_{\substack{\text{tangential component of the } \mathbf{E} \text{ field is conserved;} \\ \text{same condition as we get from Faraday's law}}} \tag{25}$$

(work this out on your own if you don't see how we got here).

- From Ampère's law, Eq. (18),

$$\tilde{H}_x^i + \tilde{H}_x^r = \tilde{H}_x^t \qquad (y \text{ components are all zero}) \tag{26}$$

$$\Rightarrow \quad \frac{E_h^i}{\eta_1}\cos\theta_i - \frac{E_h^r}{\eta_1}\cos\theta_i = \frac{E_h^t}{\eta_2}\cos\theta_t \quad \Rightarrow \quad E_h^t = \frac{\eta_2\cos\theta_i}{\eta_1\cos\theta_t}\left(E_h^i - E_h^r\right). \tag{27}$$

3.3.2 Fresnel coefficients for H polarization

- As we did last lecture, let us define the reflection and transmission coefficients

$$\rho_h = \frac{E_h^r}{E_h^i} \qquad \text{and} \qquad \tau_h = \frac{E_h^t}{E_h^i}. \tag{28}$$

- From our BCs, we have

$$1 + \rho_h = \tau_h \qquad \text{and} \qquad \frac{\eta_2\cos\theta_i}{\eta_1\cos\theta_t}(1 - \rho_h) = \tau_h. \tag{29}$$

- Solving and then applying Snell's law gives

$$\boxed{\rho_h = \frac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t} = \frac{\cos\theta_i - \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}{\cos\theta_i + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}} \tag{30}$$

and

$$\boxed{\tau_h = \frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_i + \eta_1\cos\theta_t} = \frac{2\cos\theta_i}{\cos\theta_i + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}} \tag{31}$$

where ρ_h and τ_h are the *Fresnel reflection and transmission coefficients*, respectively.

- We can see straight away that ρ_h and τ_h reduce to ρ and τ for normal incidence (Lecture 4) when $\theta_i = \theta_t = 0$.

3.4 V polarization

- To get the V-polarized Fresnel coefficients ρ_v and τ_v, simply swap $\mathbf{E}$ with $\mathbf{H}$ and $\mathbf{H}$ with $-\mathbf{E}$ (and ε with μ, so that $\eta \to 1/\eta$) and plug and chug, using the reflected-wave reference directions of Figure 3b.

- Do this exercise on your own.

- Results:

Fresnel coefficients for V polarization

$$\rho_v = \frac{E_v^r}{E_v^i} = \frac{\eta_2\cos\theta_t - \eta_1\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i} = \frac{\sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i} - (\varepsilon_2/\varepsilon_1)\cos\theta_i}{\sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i} + (\varepsilon_2/\varepsilon_1)\cos\theta_i} \tag{32}$$

$$\tau_v = \frac{E_v^t}{E_v^i} = (1 + \rho_v)\frac{\cos\theta_i}{\cos\theta_t} = \frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i} \tag{33}$$

3.5 Reflected and transmitted power

- We are often interested in the power reflected and transmitted.

  ↪ The Poynting vector gives the energy flux (power per unit area).

  ↪ From the discussion last lecture (medium 1 lossless), the magnitudes of the time-averaged power densities at the interface are

$$\mathcal{S}_h^i = \frac{|E_h^i|^2}{2|\eta_1|}\cos\phi_{\eta_1}, \qquad \mathcal{S}_h^r = \frac{|E_h^r|^2}{2|\eta_1|}\cos\phi_{\eta_1}, \qquad \mathcal{S}_h^t = \frac{|E_h^t|^2}{2|\eta_2|}\cos\phi_{\eta_2}, \tag{34}$$

  where ϕ_{η_1} is the phase of η_1 (and similarly for η_2).

  ↪ Because $\theta_i \neq \theta_t$, we need to consider the average power over an area A illuminated by the incident beam on the interface. The cross-sectional areas of the incident, reflected, and transmitted beams are

$$A_i = A_r = A\cos\theta_i \qquad \text{and} \qquad A_t = A\cos\theta_t, \tag{35}$$

$$\Rightarrow \quad P_h^i = \mathcal{S}_h^i A_i, \qquad P_h^r = \mathcal{S}_h^r A_r, \qquad P_h^t = \mathcal{S}_h^t A_t. \tag{36}$$

- Define the *reflectivity* Γ (also called *reflectance*) as the ratio of reflected power to incident power:

$$\boxed{\Gamma_h = \frac{P_h^r}{P_h^i} = \frac{|E_h^r|^2}{|E_h^i|^2} = |\rho_h|^2} \tag{37}$$

  Similarly,

$$\boxed{\Gamma_v = |\rho_v|^2} \tag{38}$$

- Define the *transmissivity* $\mathbb{T}$ (also called *transmittance*) as the ratio of transmitted power to incident power:

$$\mathbb{T}_h = \frac{P_h^t}{P_h^i} = \frac{|E_h^t|^2}{|E_h^i|^2}\,\frac{|\eta_1|\cos\theta_t\cos\phi_{\eta_2}}{|\eta_2|\cos\theta_i\cos\phi_{\eta_1}} = |\tau_h|^2\,\Phi, \qquad \mathbb{T}_v = |\tau_v|^2\,\Phi, \tag{39}$$

  where

$$\Phi = \frac{|\eta_1|\cos\theta_t\cos\phi_{\eta_2}}{|\eta_2|\cos\theta_i\cos\phi_{\eta_1}}. \tag{40}$$

  Equation (40) is exact when $\tilde{\theta}_t$ is real (lossless media with $\theta_i \leq \theta_c$, or normal incidence); for oblique incidence onto a lossy medium it is approximate (Section 4.1). Above the critical angle ($\theta_i > \theta_c$, total internal reflection), $\Gamma = 1$ and $\mathbb{T} = 0$.

4 Special considerations for oblique incidence

4.1 Lossy media

In lossy media ($\varepsilon'' > 0$, e.g., $\sigma > 0$), there will be a complex-valued angle of refraction; call it $\tilde{\theta}_t$.

★ $\tilde{\theta}_t$ is not a real angle in the traditional sense; only a real angle of transmission, θ_t (the direction normal to the planes of constant phase), has physical meaning, and in general $\theta_t \neq \mathrm{Re}(\tilde{\theta}_t)$.

- But to see how $\tilde{\theta}_t$ arises, recall that the exponential terms must be equal at the interface ($z = 0$), requiring [cf. Eq. (10)]

$$\gamma_2 \sin\tilde{\theta}_t = \gamma_1 \sin\theta_i \quad \Rightarrow \quad \sin\tilde{\theta}_t = \frac{\alpha_1 + j\beta_1}{\alpha_2 + j\beta_2}\sin\theta_i. \tag{41}$$

- $\sin\theta_i$ is always real-valued, so if either medium is lossy ($\varepsilon'' > 0$, e.g., $\sigma > 0$), the respective attenuation constant (α) is nonzero and $\tilde{\theta}_t$ is, in general, complex valued (β is always nonzero).
- In the Fresnel coefficients, Eqs. (30)–(33), θ_t is then replaced by $\tilde{\theta}_t$.

4.2 Perfect conductor, perfect reflector

- Recall that
$$\rho_h = \frac{\sqrt{\varepsilon_1}\cos\theta_i - \sqrt{\varepsilon_2}\cos\theta_t}{\sqrt{\varepsilon_1}\cos\theta_i + \sqrt{\varepsilon_2}\cos\theta_t} \qquad \text{and} \qquad \tau_h = 1 + \rho_h. \tag{42}$$
- A perfect conductor has $\varepsilon_2'' \to \infty$

  $\Rightarrow \rho_h \to -1$ and $\tau_h \to 0$ (similarly for V polarization)

  $\Rightarrow$ total, out-of-phase reflection.
- Useful for calibration, which often uses corner reflectors made of good conductors.

4.3 Brewster angle

- Recall
$$\rho_v = \frac{\sqrt{\varepsilon_1}\cos\theta_t - \sqrt{\varepsilon_2}\cos\theta_i}{\sqrt{\varepsilon_1}\cos\theta_t + \sqrt{\varepsilon_2}\cos\theta_i} \qquad \text{and} \qquad \tau_v = (1+\rho_v)\frac{\cos\theta_i}{\cos\theta_t}. \tag{43}$$
- When
$$\sqrt{\varepsilon_1}\cos\theta_t = \sqrt{\varepsilon_2}\cos\theta_i, \tag{44}$$
  $\rho_v = 0 \Rightarrow$ total transmission into medium 2 for V-polarized waves.
- Note that the equality can only hold if $\varepsilon_2/\varepsilon_1$ is real, e.g., if both materials are lossless ($\varepsilon'' = 0$); with air as medium 1, medium 2 must be lossless ($\varepsilon_2'' = 0$), not merely non-conductive.
- So, define the *Brewster angle* θ_B as the incidence angle that allows for total transmission of V-polarized EM waves:
$$\theta_B = \tan^{-1}\left(\sqrt{\frac{\varepsilon_2'}{\varepsilon_1'}}\right). \tag{45}$$
- Now, recall the H-polarized reflection coefficient
$$\rho_h = \frac{\sqrt{\varepsilon_1}\cos\theta_i - \sqrt{\varepsilon_2}\cos\theta_t}{\sqrt{\varepsilon_1}\cos\theta_i + \sqrt{\varepsilon_2}\cos\theta_t}. \tag{46}$$
  Why is there no equivalent to the Brewster angle for H polarization?

Variables

Units are SI. Values are given for universal constants; "typical" values are representative, not definitive.

Symbol	Description	Units	Value
Universal constants			
μ_0	permeability of free space	H/m	$4\pi \times 10^{-7} \approx 1.257 \times 10^{-6}$
c	speed of light in a vacuum, $1/\sqrt{\mu_0\varepsilon_0}$	m/s	2.998×10^8

Symbol	Description	Units	Value
η_0	intrinsic impedance of free space, $\sqrt{\mu_0/\varepsilon_0}$	Ω	≈ 376.7 ($\approx 120\pi$)
j	imaginary unit, $j^2 = -1$ (time dependence $e^{j\omega t}$)	–	–
Constitutive parameters			
$\varepsilon_1,\ \varepsilon_2$	complex dielectric constants (relative) of media 1 and 2, $\varepsilon' - j\varepsilon''$	dimensionless	–
$\varepsilon_1',\ \varepsilon_2'$	relative permittivities of media 1 and 2	dimensionless	≥ 1 (almost always)
ε_2''	dielectric loss factor of medium 2	dimensionless	$\to \infty$ for a perfect conductor
$\mu_1,\ \mu_2$	permeabilities of media 1 and 2	H/m	μ_0 in RS
σ	electrical conductivity	S/m	≥ 0; $\sigma > 0$ is one source of loss ($\varepsilon'' > 0$)
ω	angular frequency (set by the source)	rad/s	–
Boundary conditions			
$\mathbf{E}_1,\ \mathbf{E}_2$	(total) electric field in media 1 and 2	V/m	–
$\mathbf{H}_1,\ \mathbf{H}_2$	(total) magnetic field in media 1 and 2	A/m	–
$E^{\perp},\ H^{\perp}$	field components normal to the interface	V/m; A/m	–
$\mathbf{E}^{\parallel},\ \mathbf{H}^{\parallel}$	field components tangential to the interface	V/m; A/m	–
$\hat{\mathbf{a}}$	unit normal to the interface	dimensionless	$\hat{\mathbf{z}}$ here
Geometric optics			
$\mathbf{r}$	position vector	m	–
$x,\ y,\ z$	Cartesian coordinates	m	interface at $z = 0$
$\hat{\mathbf{x}},\ \hat{\mathbf{y}},\ \hat{\mathbf{z}}$	Cartesian unit vectors	dimensionless	–
t	time	s	–
$\mathbf{k}_i,\ \mathbf{k}_r,\ \mathbf{k}_t$	wavenumber vectors of the incident, reflected, and transmitted waves	rad/m	in the x–z plane
$k_i,\ k_r,\ k_t$	magnitudes of $\mathbf{k}_i$, $\mathbf{k}_r$, $\mathbf{k}_t$	rad/m	$k_i = k_r = k_1$
$k_{ix},\ k_{iy},\ \ldots$	Cartesian components of $\mathbf{k}_i$ (similarly $\mathbf{k}_r$, $\mathbf{k}_t$)	rad/m	$k_{iy} = k_{ry} = k_{ty} = 0$
k_1	angular wavenumber in medium 1	rad/m	k_i
$\hat{\mathbf{k}}_i,\ \hat{\mathbf{k}}_r,\ \hat{\mathbf{k}}_t$	propagation unit vectors of the incident, reflected, and transmitted waves	dimensionless	–
$u_{p1},\ u_{p2}$	phase velocities in media 1 and 2	m/s	$c/n_{1,2}'$
$n_1',\ n_2'$	real parts of the indices of refraction of media 1 and 2, $\mathrm{Re}(\sqrt{\varepsilon_{1,2}})$	dimensionless	–
$B^i,\ B^r,\ B^t$	generic coefficients of the incident, reflected, and transmitted terms in the BCs	varies	–
θ_i	angle of incidence (from the normal)	rad	–
θ_r	angle of reflection (from the normal)	rad	θ_i
θ_t	angle of refraction (transmission) (from the normal)	rad	–
θ_1	common angle of incidence and reflection, $\theta_i = \theta_r$	rad	–
θ_c	critical angle	rad	$\sin^{-1}(n_2'/n_1')$

Symbol	Description	Units	Value
Fields at oblique incidence			
$\tilde{\mathbf{E}}^i,\ \tilde{\mathbf{E}}^r,\ \tilde{\mathbf{E}}^t$	phasors of the incident, reflected, and transmitted electric fields	V/m	–
$\tilde{\mathbf{H}}^i,\ \tilde{\mathbf{H}}^r,\ \tilde{\mathbf{H}}^t$	phasors of the incident, reflected, and transmitted magnetic fields	A/m	–
$\tilde{E}_x^i,\ \tilde{E}_y^i,\ \tilde{E}_z^i$	Cartesian components of $\tilde{\mathbf{E}}^i(\mathbf{0})$ (similarly for r, t, and $\tilde{\mathbf{H}}$)	V/m; A/m	–
$\tilde{\mathbf{E}}_h^i,\ \tilde{\mathbf{E}}_h^r,\ \tilde{\mathbf{E}}_h^t$	phasors of the H-polarized fields (similarly $\tilde{\mathbf{H}}_h$; V polarization: subscript v)	V/m	–
$E_h^i,\ E_h^r,\ E_h^t$	complex amplitudes of the H-polarized $\mathbf{E}$ fields at the origin	V/m	E_h^i known
$E_v^i,\ E_v^r,\ E_v^t$	complex amplitudes of the V-polarized $\mathbf{E}$ fields at the origin	V/m	E_v^i known
$\gamma_1,\ \gamma_2$	propagation constants of media 1 and 2, $\alpha + j\beta$	m^{-1}	–
$\alpha_1,\ \alpha_2$	attenuation constants of media 1 and 2	Np/m	0 if lossless
$\beta_1,\ \beta_2$	phase constants of media 1 and 2	rad/m	–
$\eta_1,\ \eta_2$	intrinsic impedances of media 1 and 2, $\eta_0/\sqrt{\varepsilon_{1,2}}$	Ω	real if lossless
$\phi_{\eta_1},\ \phi_{\eta_2}$	phase angles of η_1 and η_2	rad	0 if lossless
Fresnel coefficients and power			
$\rho,\ \tau$	Fresnel reflection and transmission coefficients at normal incidence (Lecture 4)	dimensionless	–
$\rho_h,\ \rho_v$	Fresnel reflection coefficients for H and V polarization	dimensionless	ρ at $\theta_i = 0$
$\tau_h,\ \tau_v$	Fresnel transmission coefficients for H and V polarization	dimensionless	τ at $\theta_i = 0$
$\mathcal{S}_h^i,\ \mathcal{S}_h^r,\ \mathcal{S}_h^t$	magnitudes of the time-averaged power densities of the H-polarized waves	$\mathrm{W/m}^2$	–
A	area illuminated by the incident beam on the interface	m^2	–
$A_i,\ A_r,\ A_t$	cross-sectional areas of the incident, reflected, and transmitted beams	m^2	$A\cos\theta_i$, $A\cos\theta_i$, $A\cos\theta_t$
$P_h^i,\ P_h^r,\ P_h^t$	average powers of the incident, reflected, and transmitted H-polarized beams	W	–
$\Gamma_h,\ \Gamma_v$	reflectivities for H and V polarization, $\lvert\rho_{h,v}\rvert^2$	dimensionless	0–1
$\mathbb{T}_h,\ \mathbb{T}_v$	transmissivities for H and V polarization, $\lvert\tau_{h,v}\rvert^2\Phi$	dimensionless	0–1
Φ	factor relating $\mathbb{T}$ to $\lvert\tau\rvert^2$	dimensionless	–
Special considerations			
$\tilde{\theta}_t$	complex-valued angle of refraction (lossy media)	rad	real if $\varepsilon_2/\varepsilon_1$ is real and $\theta_i \le \theta_c$
θ_B	Brewster angle	rad	$\tan^{-1}\sqrt{\varepsilon_2'/\varepsilon_1'}$